

PART A (25 MARKS)

This section consists of **3** questions. Answer **all** questions.

1. a) The function $g(x)$ is defined by

$$g(x) = \begin{cases} 2 & , \quad x < 1 \\ \frac{x-2}{\sqrt{2x-2}} & , \quad x = 1 \\ \frac{|8-x|}{x-8} & , \quad x > 1 \end{cases}$$

Find

i) $\lim_{x \rightarrow 2^+} g(x)$. **[3 marks]**

ii) $\lim_{x \rightarrow 8^+} g(x)$. **[3 marks]**

b) Evaluate $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$. **[3 marks]**

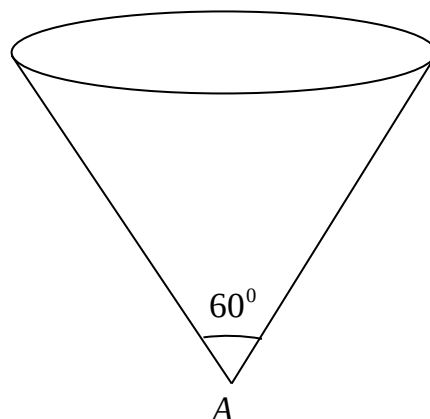
2. The parametric equations of a curve is given by

$$x = e^{2t+1}, \quad y = e^{-(2t-1)}.$$

a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $t = 1$. **[7 marks]**

b) Given $z = x^2 - xy$. Express z in terms of t and find $\frac{dz}{dt}$. **[4 marks]**

3. A cone-shaped tank as shown below.



Water flows through a hole A at rate of 6 cm^3 per second. Find the rate of change in height of the water when the volume of water in the cone is $24\pi \text{ cm}^3$.

[5 marks]

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. Given a complex number $z = \frac{2i}{\sqrt{3}-i}$.

a) State z in the form of $a + bi$, where a and b are real numbers. [5 marks]

b) Find the modulus and argument of z . Hence express in the polar form.

[5 marks]

2 a) Solve the inequality $\left| \frac{3}{x-4} \right| = 7$, $x \neq 4$ [5 marks]

b) Find the value of x which satisfies the equation

$$\log_9 x = (\log_3 x)^2, \quad x > 1. \quad [6 \text{ marks}]$$

3 Determine whether $f(x) = \frac{1}{x-4}$ and $g(x) = \frac{4x+1}{x}$ are inverse function of each other by computing their composite functions.

[6 marks]

4 Given $g(x) = e^{-x} + 2$.

i) Show that f is one to one function [3 marks]

ii) Find $g^{-1}(x)$, and state its domain and range. [6 marks]

iii) Hence, sketch the graph g and g^{-1} on the same axes. [3 marks]

5 a) Given
$$f(x) = \begin{cases} 5 - px & , \quad -2 < x \leq -1 \\ x^2 + px + q & , \quad -1 < x \leq 2 \\ \frac{x^2 - 4}{x - 2} & , \quad x > 2 \end{cases}$$

Find the values of p and q if $f(x)$ is continuous for all real values of x .

[7 marks]

b)
$$\lim_{x \rightarrow 3} \frac{2 - \sqrt{x+1}}{x^2 - 9}$$

[4 marks]

6 a) If f is a function with $f'(1) = 2$, find $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{\sqrt{x} - 1}$.

[7 marks]

b) i) Find $\frac{dy}{dx}$ if $y = \operatorname{cosec}\{\sin[\ln(x+1)]\}$.

[4 marks]

ii) Obtain the second derivative of $y = \frac{\cos 3x}{e^{2x}}$ and express your answer

in the simplest form.

[5 marks]

7 Given a function $f(x) = x^4 - 18x^2 + 32$

a) Find the values of x such that $f'(x) = 0$

[3 marks]

b) By using the second derivative test, determine the maximum and minimum points of $f(x)$

[6 marks]

END OF QUESTION PAPER

SUGGESTED ANSWER**PART A**

1 a i) 2

ii) 1

b) -1

2 a) $\frac{dy}{dx} = -\frac{1}{e^4}$; $\frac{d^2y}{dx^2} = \frac{2}{e^7}$

b) $z = e^{4t+2} - e^2$; $\frac{dz}{dt} = 4e^{4t+2}$

3. $\frac{dh}{dt} = -\frac{1}{2\pi} \text{ cm s}^{-1}$

PART B

1. a) $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

b) $|z| = 1$; $\text{Arg}(z) = 2.094 \text{ rad}$

$z = 1(\cos 2.094 + i \sin 2.094)$

2. a) $x = \frac{25}{7}$ or $x = \frac{31}{7}$

b) $x = \sqrt{3}$

3. $f(x)$ and $g(x)$ are inverse function of each other

4. ii) $g^{-1}(x) = -\ln(x-2)$ $D_{g^{-1}} = (2, \infty)$ $R_{g^{-1}} = (-\infty, \infty)$

5. a) $p = -1$, $q = 2$

b) $-\frac{1}{24}$

6. a) 4

b) i) $\frac{dy}{dx} = \frac{-\cos[\ln(x+1)]}{x+1} \cos ec[\sin(\ln(x+1))] \cot[\sin(\ln(x+1))]$

ii) $\frac{d^2y}{dx^2} = e^{-2x}(12 \sin 3x - 5 \cos 3x)$

7. a) $x = 0$, $x = 3$, $x = -3$

b) Maximum point $(0, 32)$; Minimum points $(-3, -49)$ and $(3, -49)$